

Communication Electronics Cheat Sheet (v2)

1. Waves & Antennas

- Wavelength:

$$\lambda = \frac{c}{f}$$

- Quarter-wave antenna:

$$L = \lambda/4$$

- Half-wave dipole:

$$L = \lambda/2$$

- Propagation time:

$$t = d/v_p$$

- Velocity in coax:

$$v = c \cdot VF$$

2. Decibels & Gains

- Voltage gain (dB):

$$G_v(dB) = 20 \log_{10}(V_{out}/V_{in})$$

- Current gain (dB):

$$G_i(dB) = 20 \log_{10}(I_{out}/I_{in})$$

- Power gain (dB):

$$G_p(dB) = 10 \log_{10}(P_{out}/P_{in})$$

- Cascaded gains:

$$G_{tot} = \prod A_n, \quad G_{dB} = \sum G_{n,dB}$$

- Power in dBm:

$$P_{dBm} = 10 \log_{10}(P/1mW)$$

3. RC Filters

- Corner frequency:

$$f_c = \frac{1}{2\pi RC}$$

- Attenuation slopes:

- Single pole: -20 dB/decade (−6 dB/octave)
- Two pole: -40 dB/decade (−12 dB/octave)

- Transfer function:

$$H(j\omega) = \frac{\omega_c}{j\omega + \omega_c}$$

- At corner frequency:

$$|H(j\omega_c)| = 1/\sqrt{2} = -3dB$$

4. Resonance & RLC Circuits

- Inductive reactance:

$$X_L = 2\pi fL$$

- Capacitive reactance:

$$X_C = \frac{1}{2\pi fC}$$

- Resonant frequency:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

- Quality factor:

$$Q = \frac{X_L}{R_s} = \frac{f_0}{BW}$$

- Bandwidth:

$$BW = f_0/Q$$

5. Sallen-Key Filters

- Component relations:

$$C_2 = 2C_1$$

$$R_1 = R_2 = \frac{1}{2\pi f_c C_1 \sqrt{2}}$$

6. Oscillators & AM

- Oscillator condition (Barkhausen):

$$|A\beta| \geq 1, \quad \angle A\beta = 0^\circ$$

- AM Signal:

$$s(t) = [1 + m \cos(\omega_m t)] \cos(\omega_c t)$$

- Modulation index:

$$m = A_m/A_c$$